Appendix for ECS Poster

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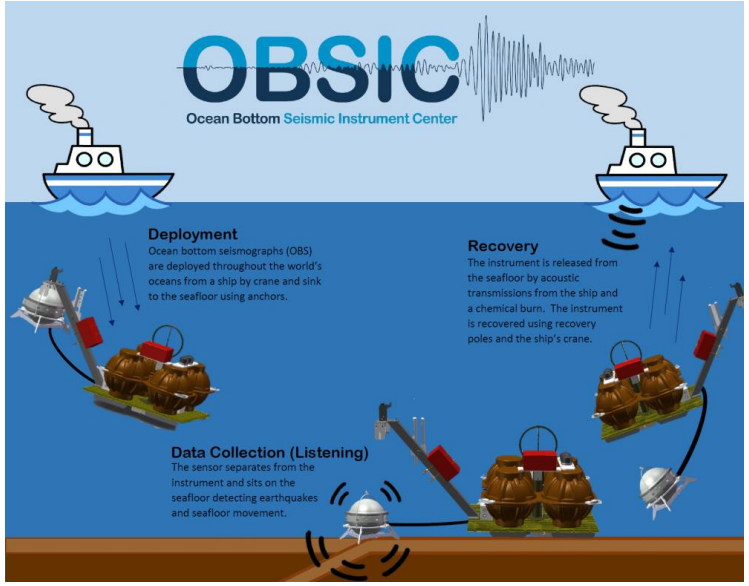
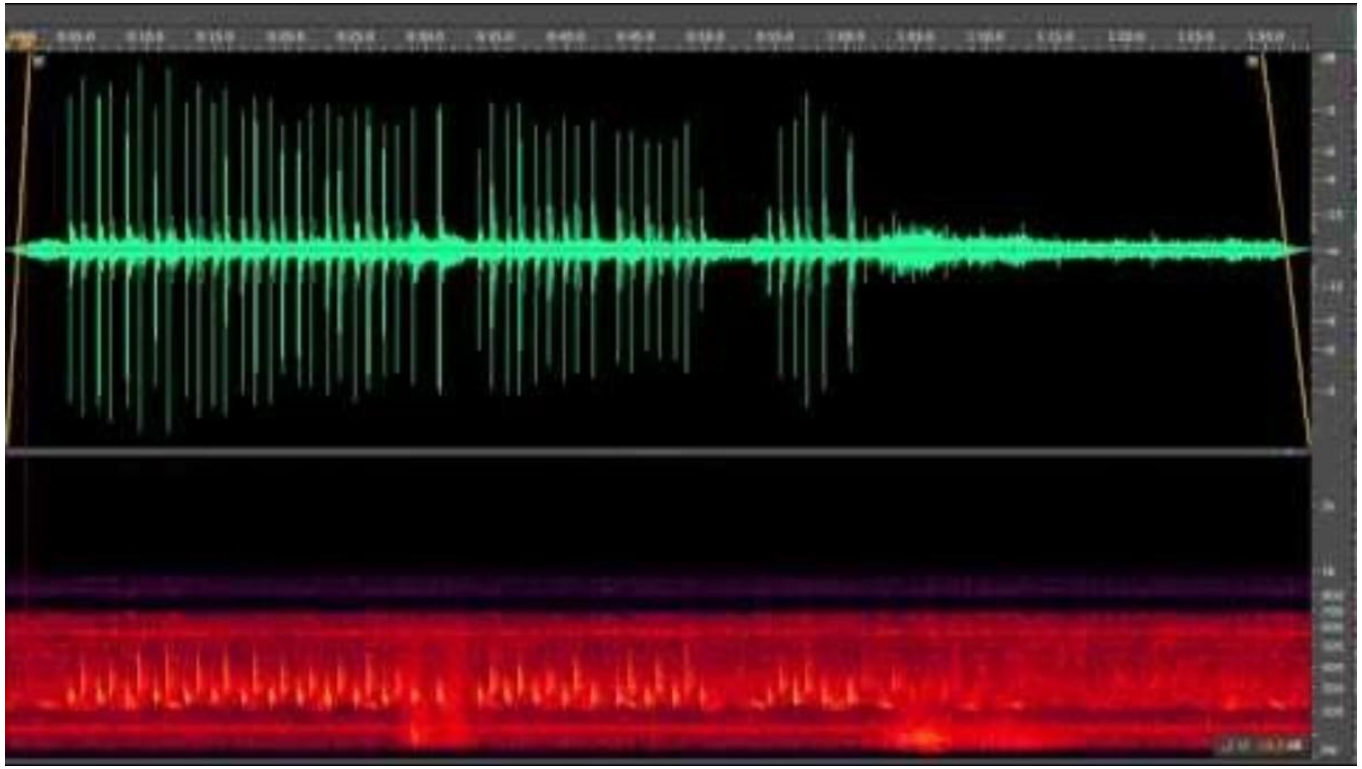
What is an Ocean Bottom Seismometer (OBS)?



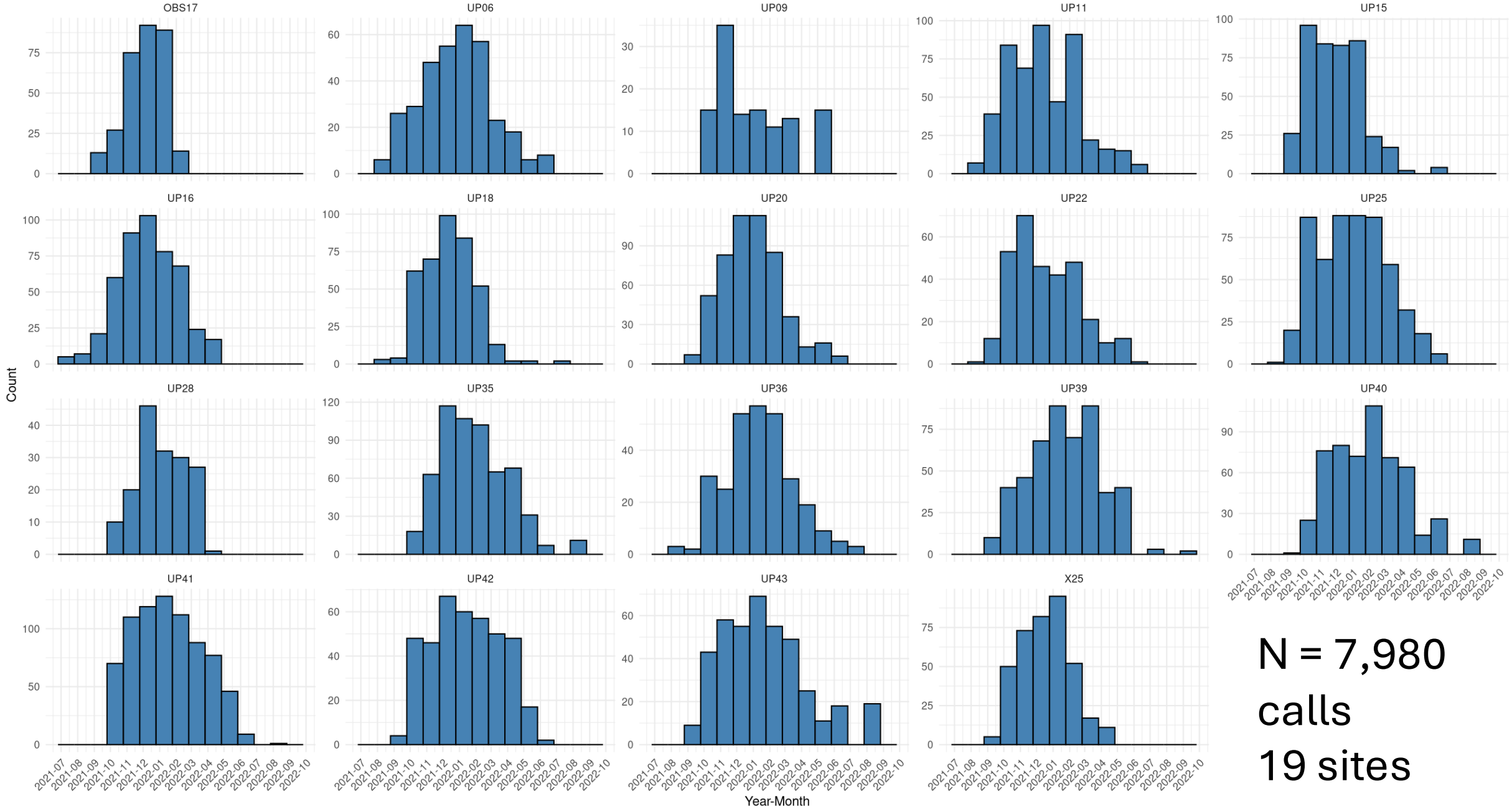
(WHOI, 2022)

- **Main objective:** understand the structure of the Earth's crust by measuring ground motion (e.g. earthquakes)
- **Common configuration:** 3-component OBS (4 sensors – 3 geophones + 1 hydrophone)
- **Type of array:** cabled or autonomous
- **Used by:** researchers, government agencies, commercial companies
- Passive or active source surveys (i.e. air guns)
- Some datasets are publicly available (i.e. through IRIS)

Example of Fin Whale Calls in Seismic Data



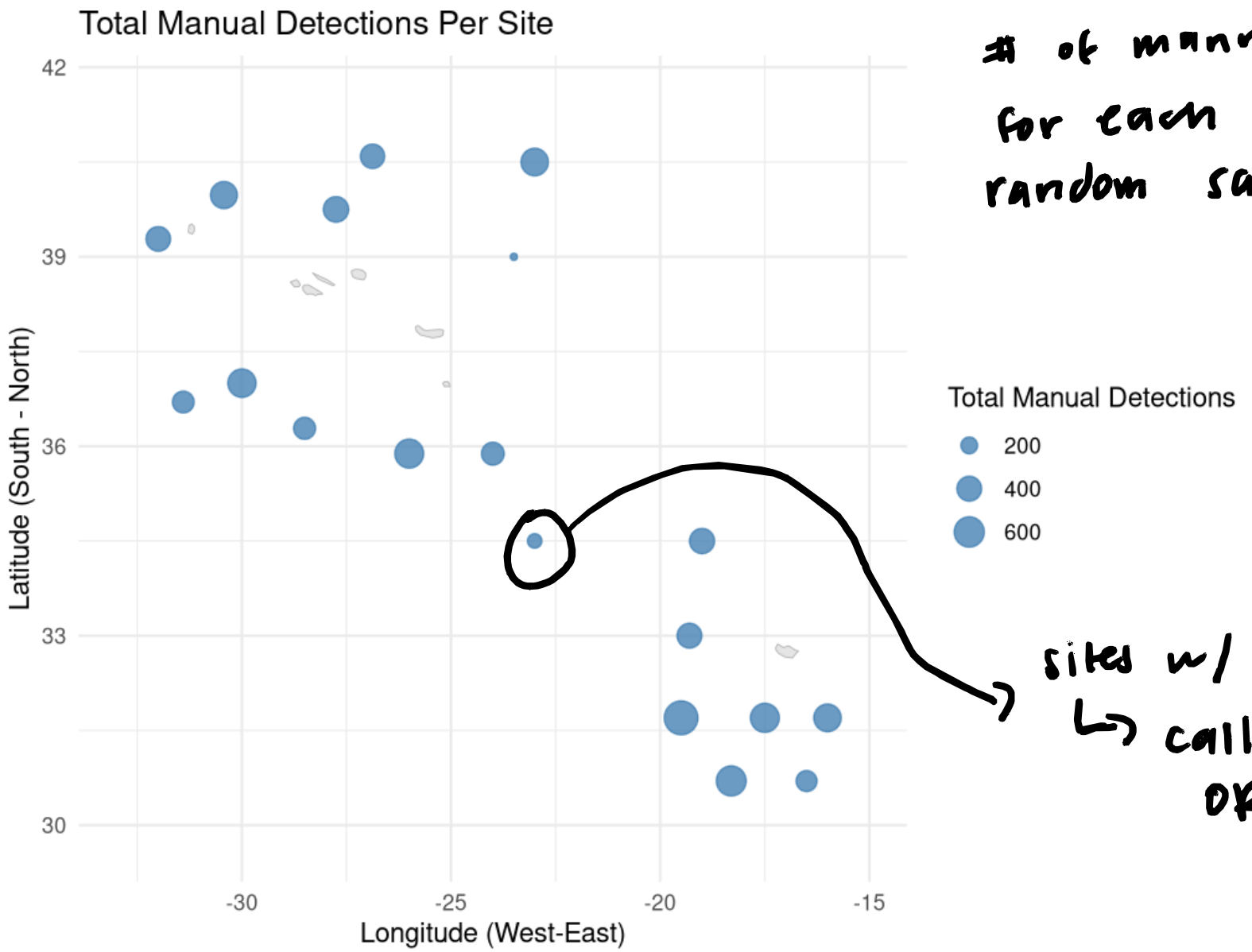
Manual Detections – Temporal Variability



N = 7,980 calls

19 sites

Manual Detections – Spatial Variability



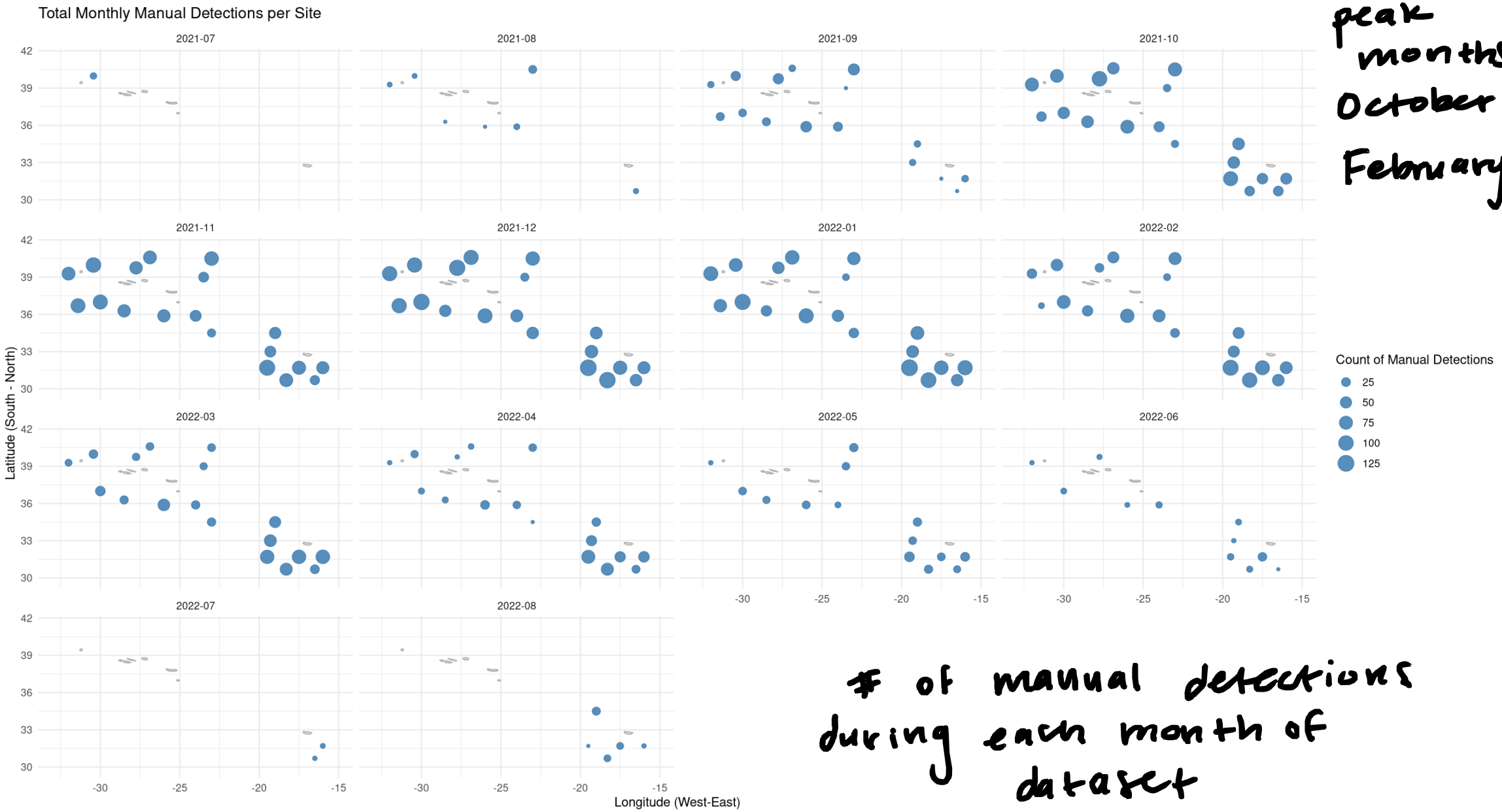
# of manual detections for each site in the random sample!

↓

same amount of time annotated for each site

sites w/ less detections
↳ calls less detectable
OR less calls?

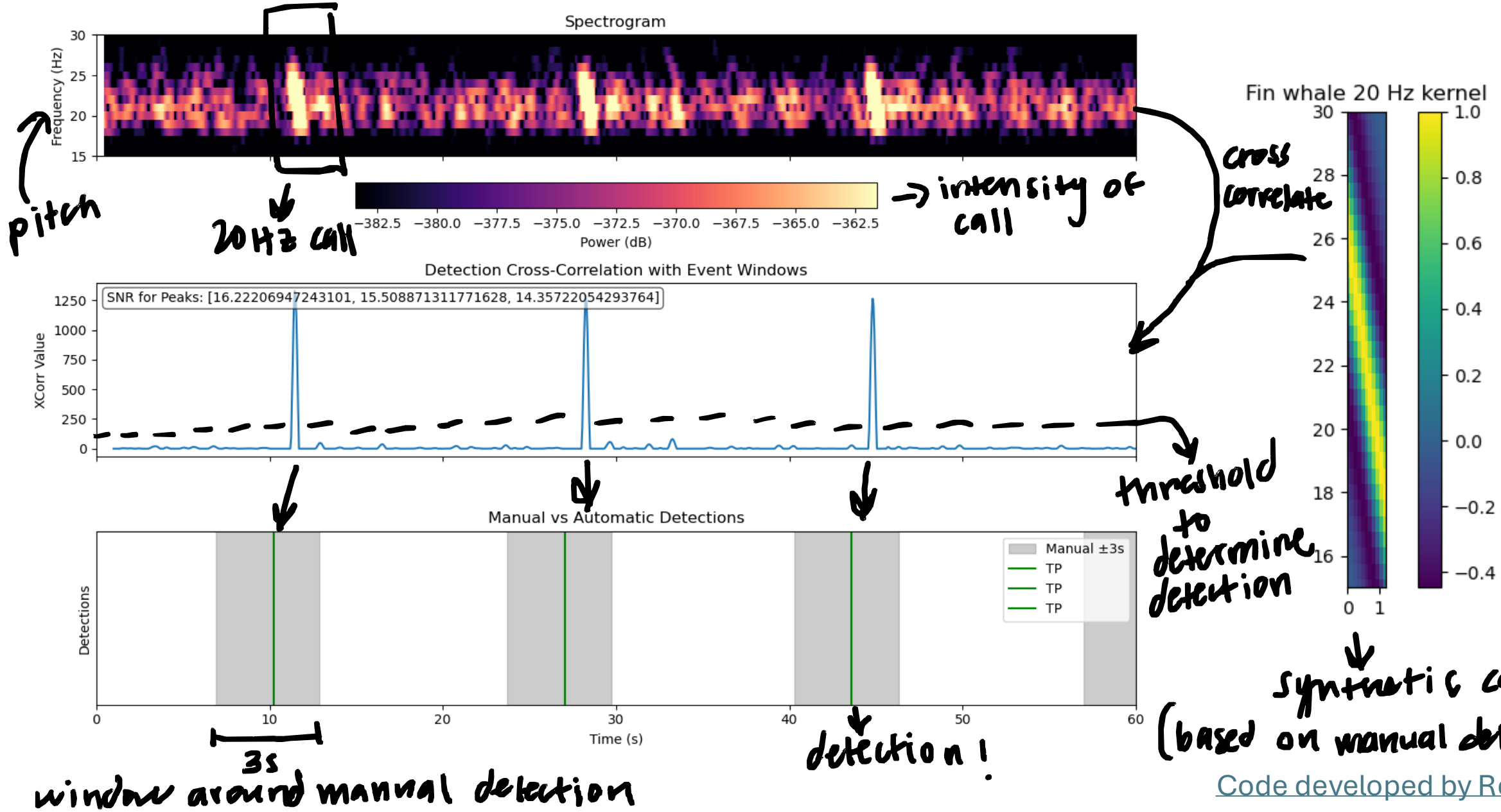
Manual Detections – Spatial and Temporal Variability



peak months October - February

# of manual detections during each month of dataset

Automatic Detector – Spectrogram Cross Correlation

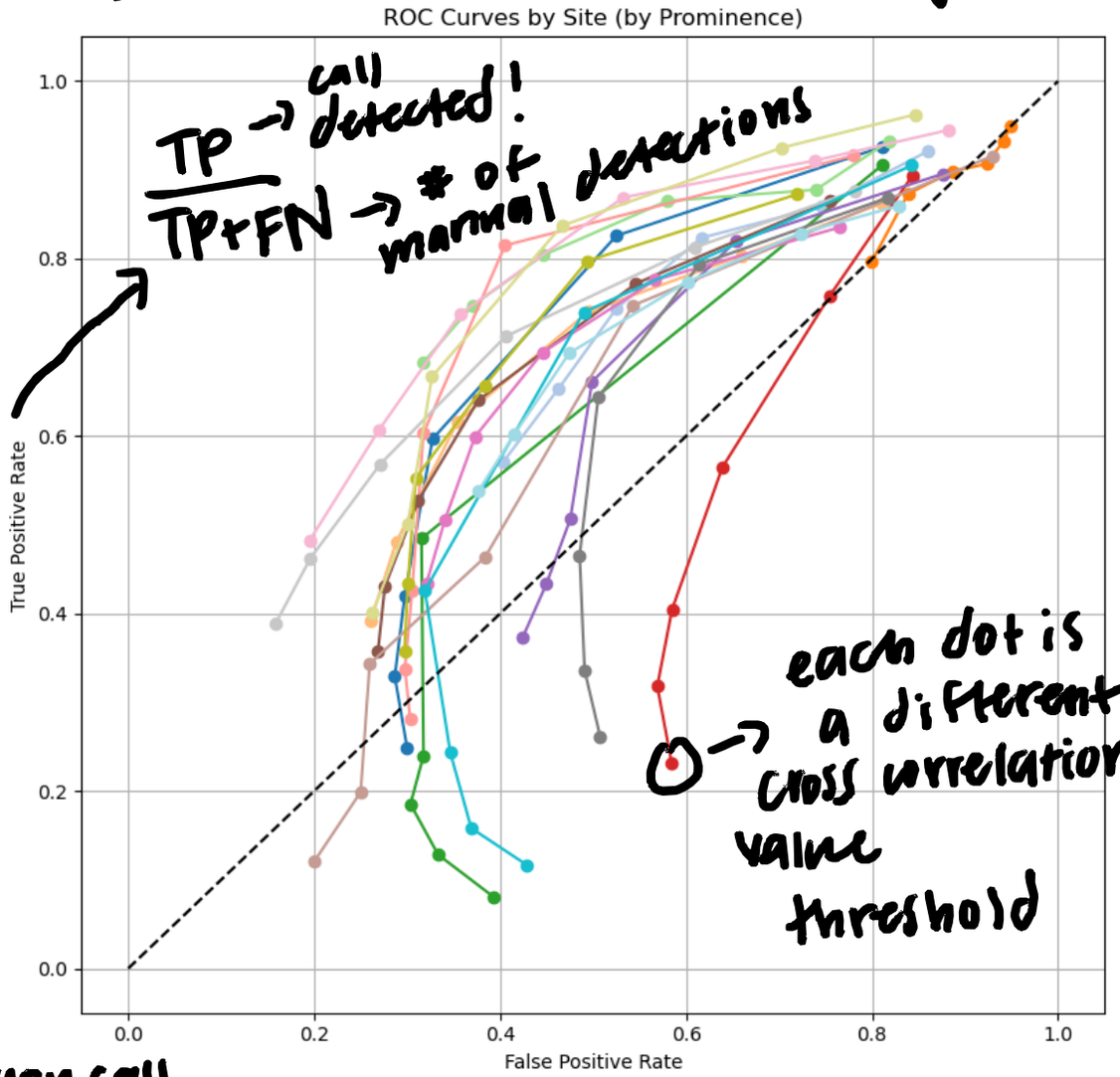


Code developed by Rose Hilmo

Sites Perform Differently !!!!

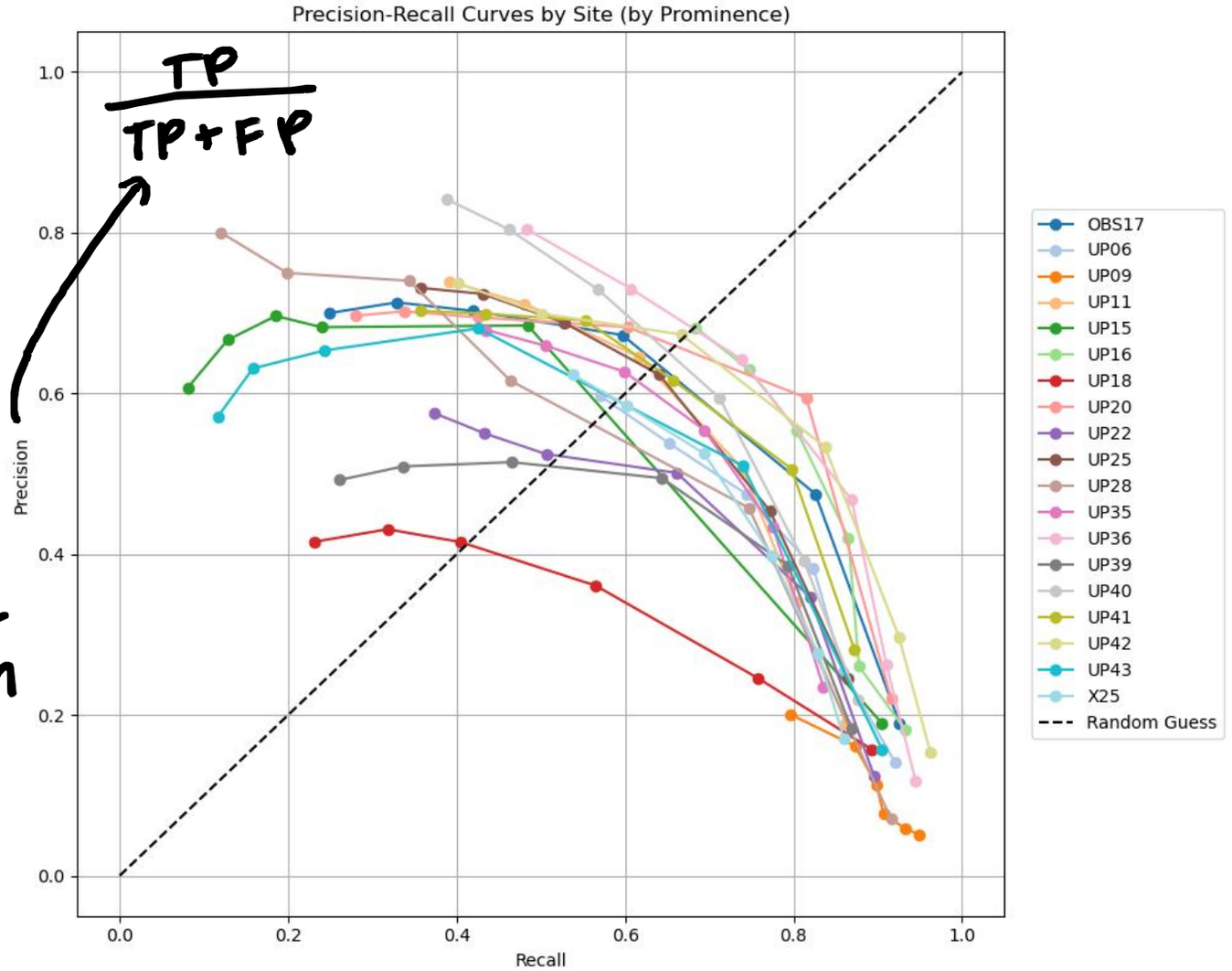
w/ same detector inputs + synthetic call

trade off b/w # of true positives and false positives!



$\frac{TP}{TP+FN}$ → call detected! → # of manual detections

each dot is a different cross correlation value threshold

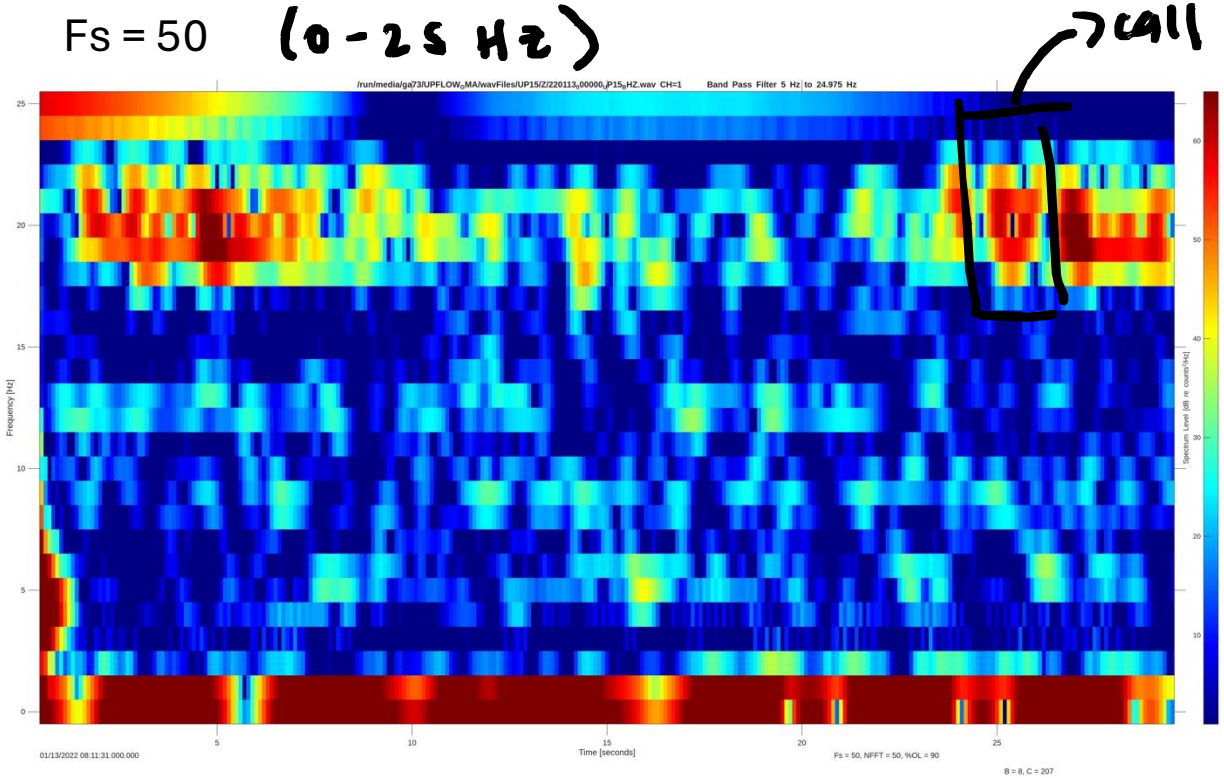


$\frac{TP}{TP+FP}$

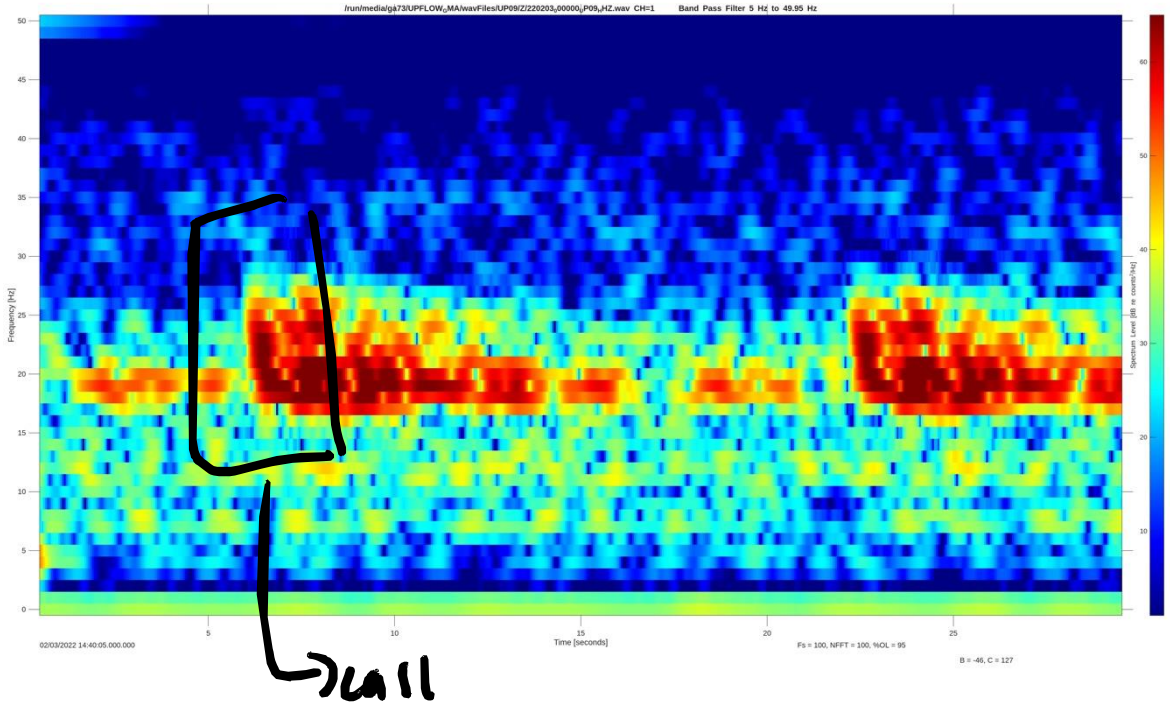
non call detected ← FP / (FP + TP) → total # of detections

true positive rate

Fs = 50 (0-25 Hz)



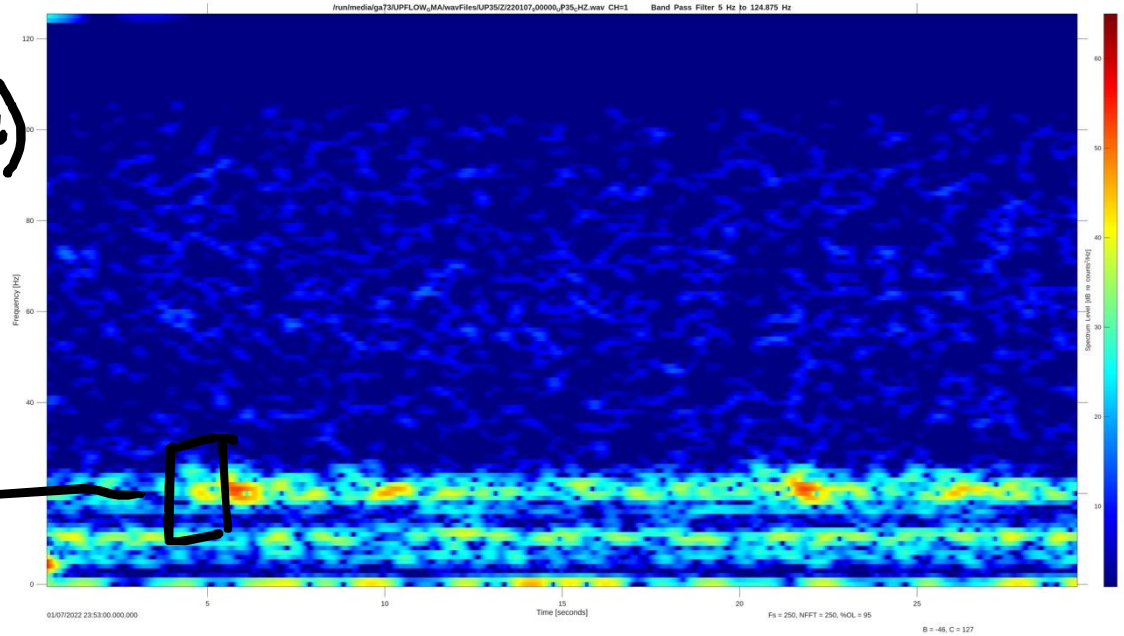
Fs = 100 (0-50 Hz)



can see full bandwidth of call!

More examples of how the calls look like with different sampling rates (Fs)

Fs = 250 (0-125 Hz)

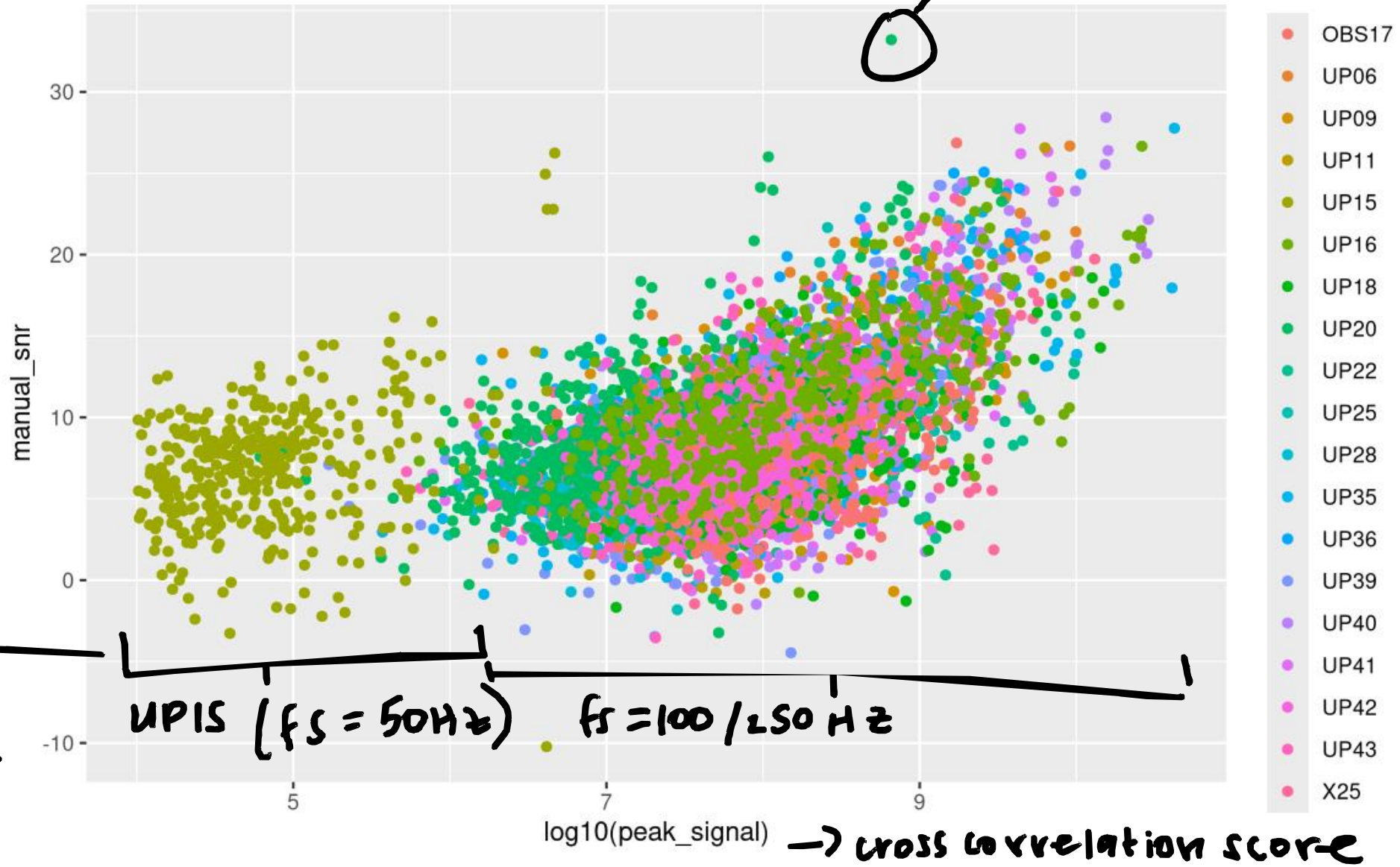


50 Hz Site vs the Other Sampling Rates – Cross Correlation Score

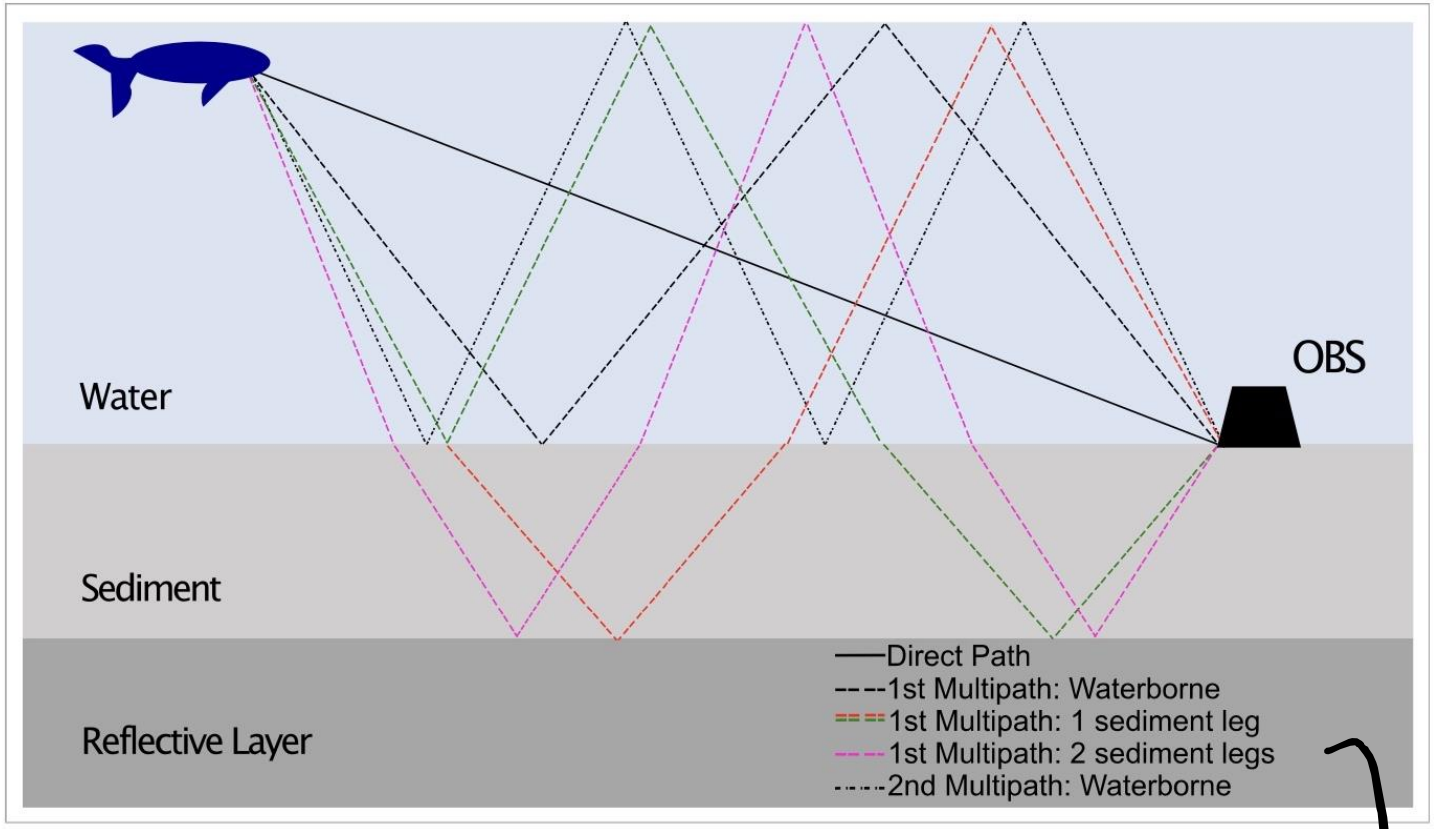
each dot is a true positive

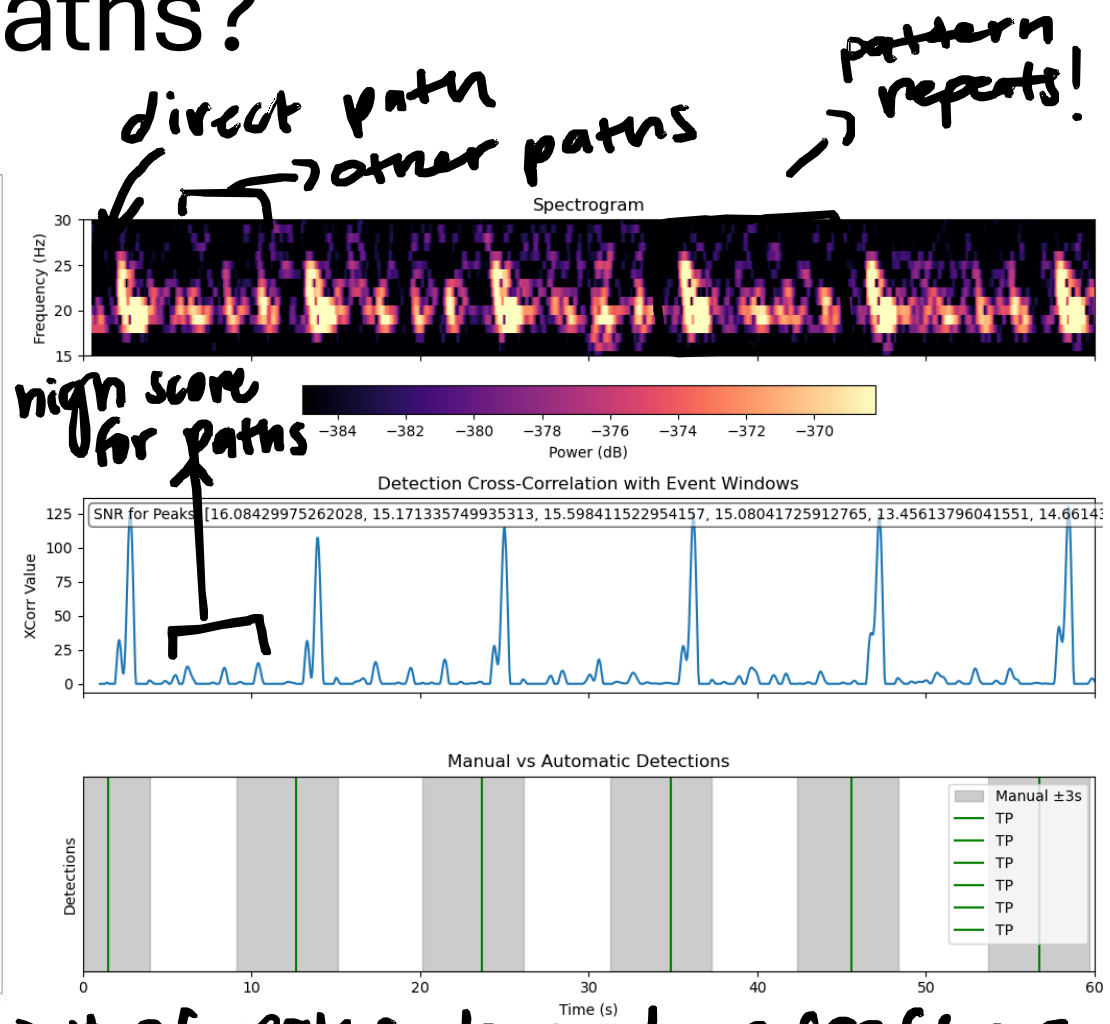
signal to noise ratio (dB)

lower scores for 50 Hz calls when same detector is use



What are call echoes/multipaths?





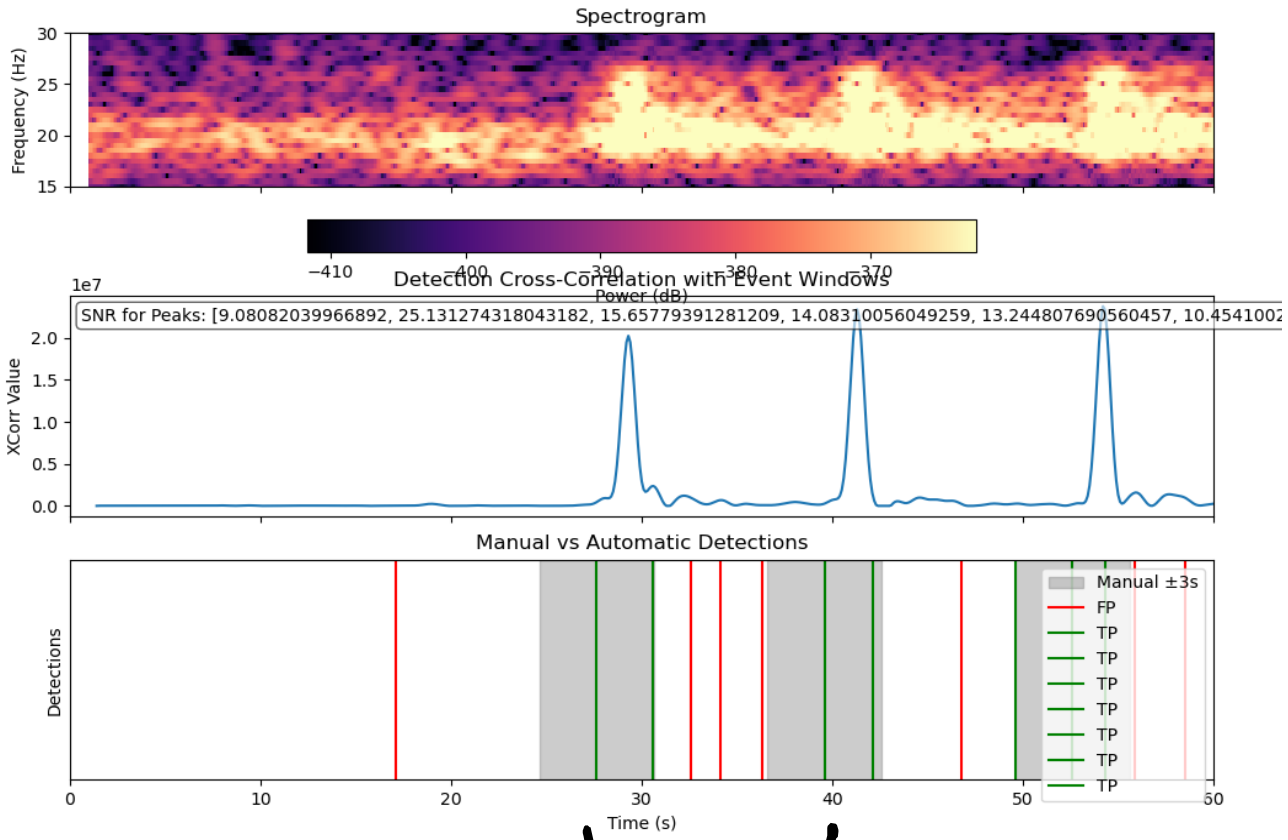
fin whale is calling @ the surface while OBS is at the seafloor

↳ many paths that the sound takes to the OBS

↳ # of paths depend on seafloor characteristics + water properties

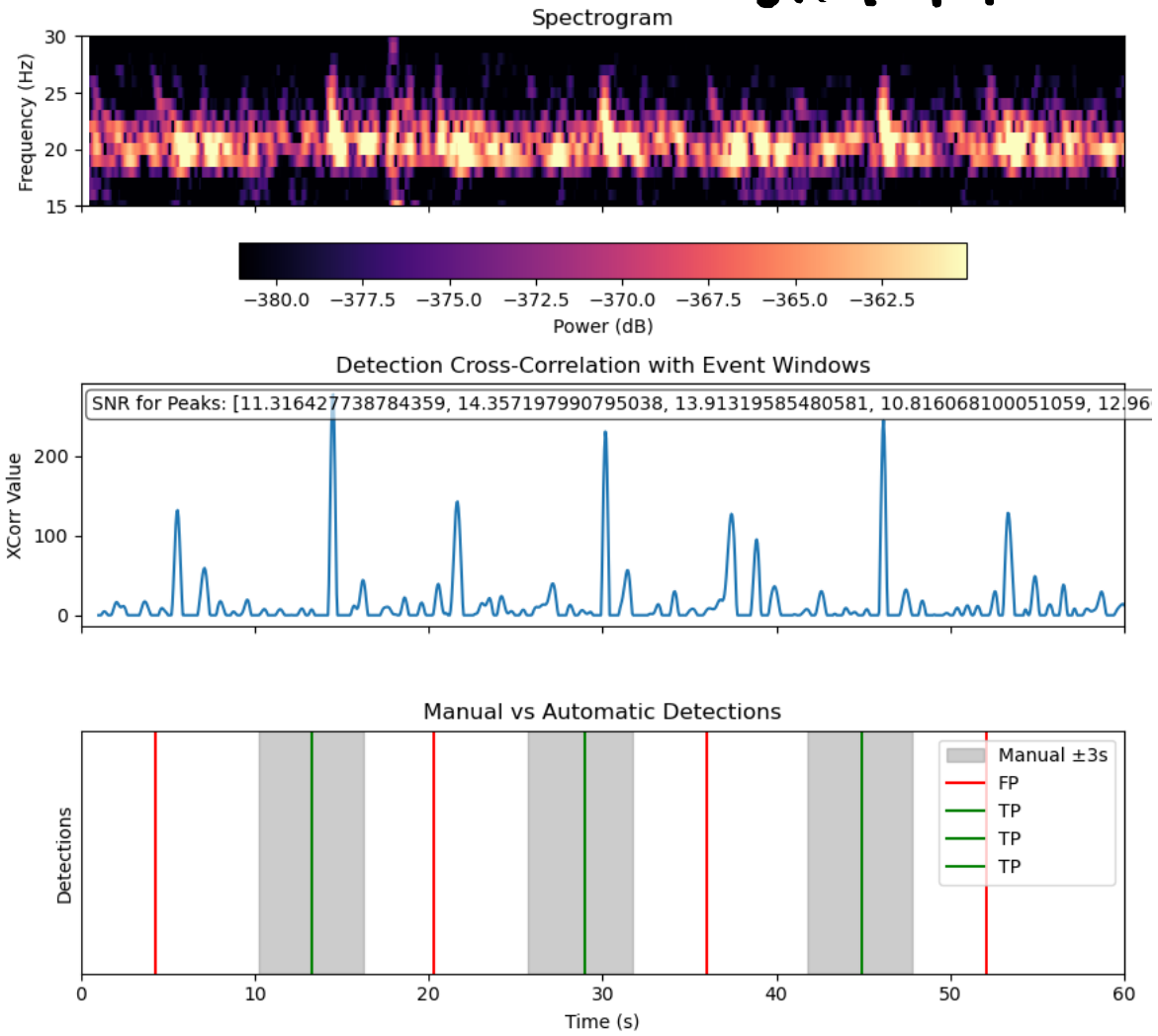
Other Complications – False Positives Caused by Paths

Buffer = 1s → time between detections



multipath/echoes detected ↑ FP

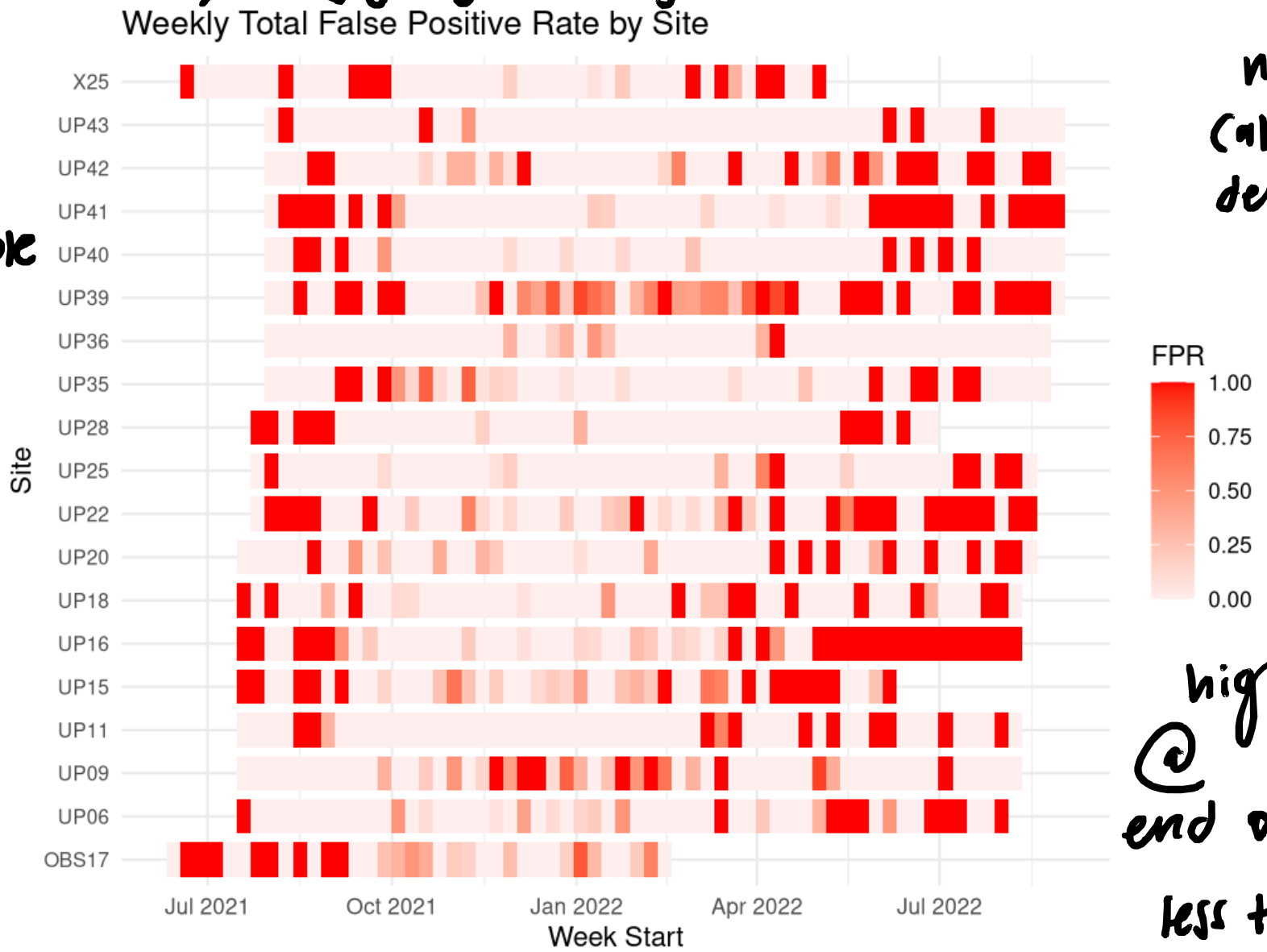
Buffer = 7s → less paths detected but still ↑ FP



False Positive Rate Using Manual Detections

visual inspection of detections in manual subsample to annotate TP + FP

aggregated by week



FPR necessary for calculation of density

↓

varies per site + season

↓

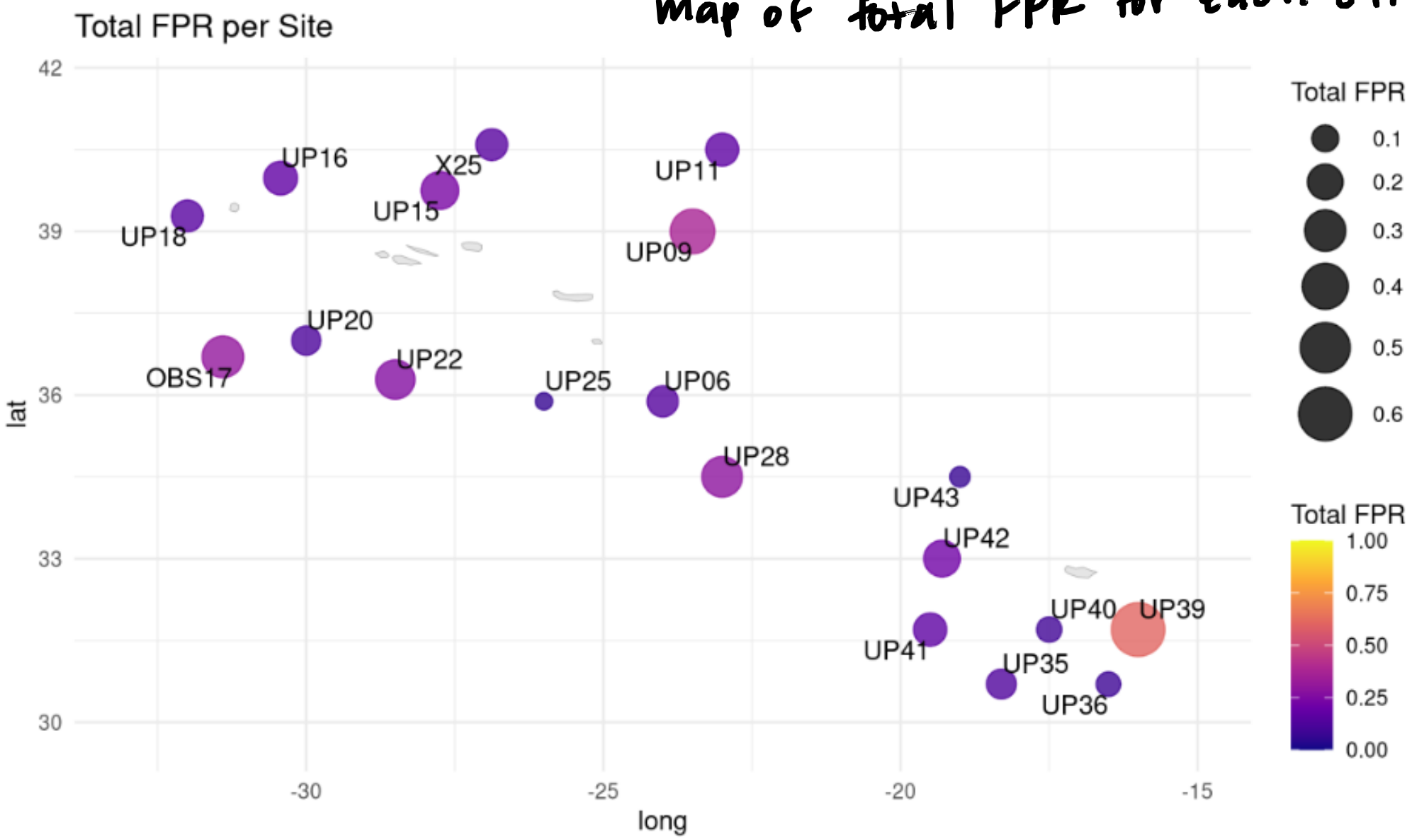
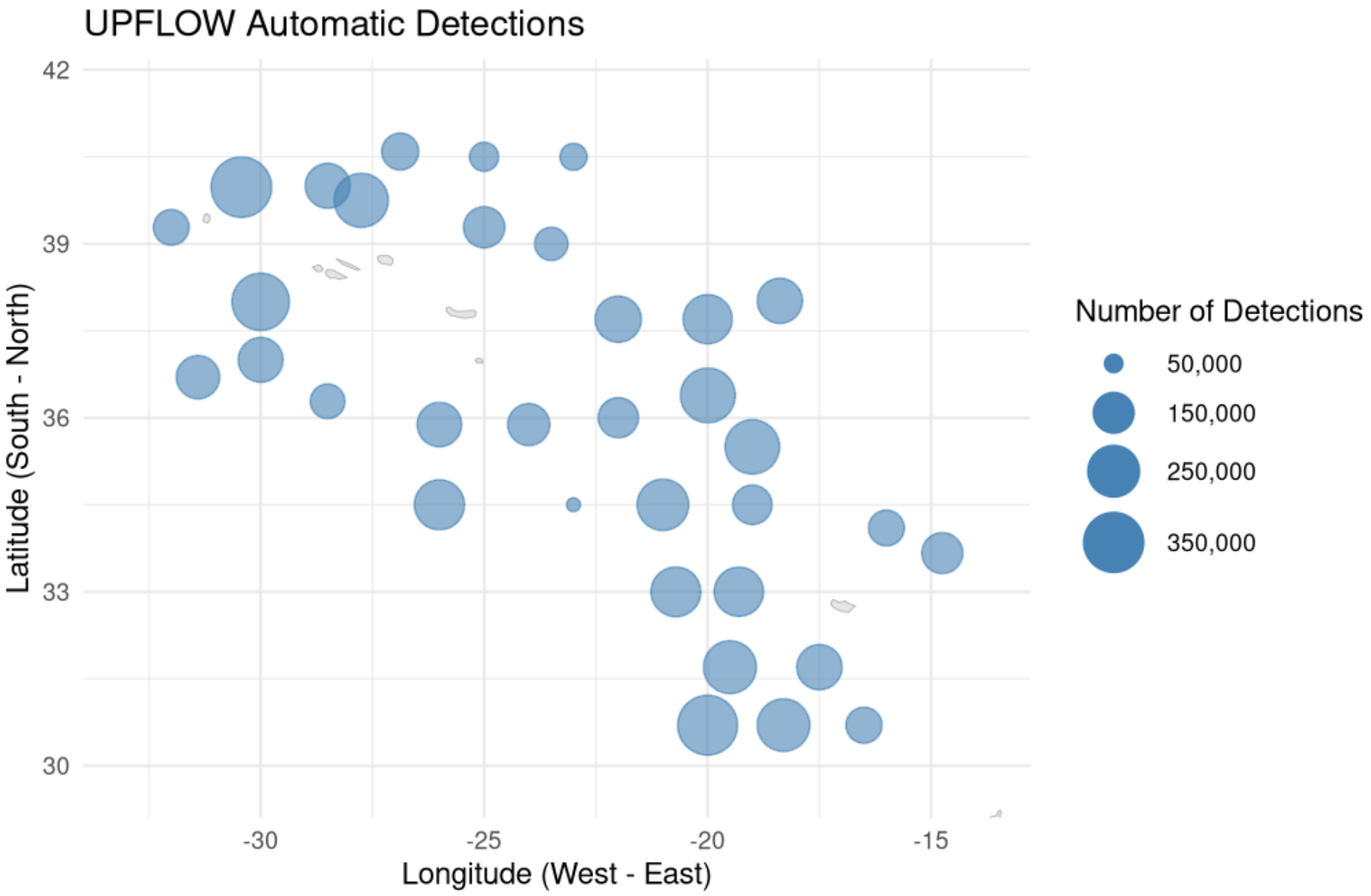
higher FPR @ start and end of deployment

↓

less true positives!

False Positive Rate Using Manual Detections

map of total FPR for each site



Automatic Detections → detector applied to entire dataset!